



A short and simple proof of Ramanujan's mod 11 partition congruence

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Abstract

We present a short and simple proof of Ramanujan's partition congruence $p(11n + 6) \equiv 0 \pmod{11}$, and dedicate it to him on the occasion of his 125th birthday.

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1. Introduction

Let $p(n)$ be the number of partitions of n . It was known to Euler that

$$\sum_{n \geq 0} p(n)q^n = \frac{1}{E(q)},$$

where

$$E(q) = \prod_{n \geq 1} (1 - q^n) = \sum_{n=-\infty}^{\infty} (-1)^n q^{(3n^2-n)/2} = 1 - q - q^2 + q^5 + q^7 - \dots$$

It follows that for $n \geq 1$,

$$p(n) = p(n-1) + p(n-2) - p(n-5) - p(n-7) + \dots$$

This recurrence can be used to calculate $p(n)$. P. A. MacMahon did this in 1917 for $n \leq 200$, and Ramanujan made the observations, among others, that

$$p(5n + 4) \equiv 0 \pmod{5}, \quad p(7n + 5) \equiv 0 \pmod{7} \quad \text{and} \quad p(11n + 6) \equiv 0 \pmod{11}.$$

It is possible to use an identity of Jacobi, namely

$$E(q)^3 = \sum_{n \geq 0} (-1)^n (2n+1) q^{(n^2+n)/2}$$

to give really simple proofs of the mod 5 and mod 7 congruences.

Thus, modulo 5,

$$E(q)^3 \equiv J_0 + J_1,$$

where J_i consists of terms in which the power of q is congruent to i modulo 5, and so, modulo 5,

$$\begin{aligned}\sum_{n \geq 0} p(n)q^n &= \frac{1}{E(q)} = \frac{(E(q)^3)^3}{(E(q)^5)^2} \equiv \frac{(E(q)^3)^3}{E(q^5)^2} \equiv \frac{(J_0 + J_1)^3}{E(q^5)^2} \\ &= \frac{J_0^3 + 3J_0^2J_1 + 3J_0J_1^2 + J_1^3}{E(q^5)^2}.\end{aligned}$$

If we extract those terms in which the power of q is $4 \pmod{5}$, we find

$$\sum_{n \geq 0} p(5n+4)q^{5n+4} \equiv 0$$

and

$$p(5n+4) \equiv 0.$$

Modulo 7,

$$E(q)^3 \equiv J_0 + J_1 + J_3,$$

where J_i consists of terms in which the power of q is congruent to i modulo 7, and so

$$\begin{aligned}\sum_{n \geq 0} p(n)q^n &= \frac{1}{E(q)} = \frac{(E(q)^3)^2}{E(q)^7} \equiv \frac{(E(q)^3)^2}{E(q^7)} \equiv \frac{(J_0 + J_1 + J_3)^2}{E(q^7)} \\ &\equiv \frac{J_0^2 + 2J_0J_1 + J_1^2 + 2J_0J_3 + 2J_1J_3 + J_3^2}{E(q^7)}\end{aligned}$$

and consequently

$$p(7n+5) \equiv 0 \pmod{7}.$$

It has long been believed that a similar proof of the mod 11 congruence is not possible. I intend to demonstrate that this is not the case, and I present this proof to Ramanujan on the occasion of the 125th anniversary of his birth.

Proofs of the mod 11 congruence have previously been given by Ramanujan [15], Paper 30, Atkin and Swinnerton-Dyer [3], Winquist [16], Garvan [6], Hirschhorn [10], Garvan [7], Garvan and Stanton [9], Hirschhorn [11], Hirschhorn [12], Ekin [5], Hirschhorn [13], Ramanujan [1], p. 106 and Marivani [14].

Apart from these, we have all the proofs given by the crank, for which see Dyson [4], Garvan [7], Andrews and Garvan [2] and Garvan, Kim and Stanton [8], just to list the first few.

But perhaps none of these is as simple or direct as the proof given here.

2. The proof

Modulo 11, we have

$$(E(q)^3)^4 = E(q)^{12} = E(q)^{11}E(q) \equiv E(q^{11})E(q).$$

Now,

$$E(q) = E_0 + E_1 + E_2 + E_4 + E_5 + E_7,$$

where E_i consists of terms in which the power of q is congruent to i modulo 11 and

$$E(q)^3 \equiv J_0 + J_1 + J_3 + J_6 + J_{10},$$

where J_i consists of terms in which the power of q is congruent to i modulo 11.

So

$$(J_0 + J_1 + J_3 + J_6 + J_{10})^4 \equiv E(q^{11})(E_0 + E_1 + E_2 + E_4 + E_5 + E_7).$$

If we consider powers of q congruent to 3, 6, 8, 9 and 10, we find that, modulo 11,

$$\begin{aligned} J_0^3 J_3 + J_0 J_1^3 + 6J_0 J_1 J_3 J_{10} + 7J_1^2 J_6^2 + 3J_3 J_6^2 J_{10} + J_6 J_{10}^3 &\equiv 0, \\ J_0^3 J_6 + 7J_0^2 J_3^2 + 6J_0 J_1 J_6 J_{10} + J_1^3 J_3 + 3J_1 J_3^2 J_{10} + J_6^3 J_{10} &\equiv 0, \\ 3J_0 J_1^2 J_6 + 6J_0 J_3 J_6 J_{10} + J_0 J_{10}^3 + 7J_1^2 J_3^2 + J_1 J_6^3 + J_3^3 J_{10} &\equiv 0, \\ 3J_0^2 J_3 J_6 + 7J_0^2 J_{10}^2 + J_0 J_3^3 + 6J_1 J_3 J_6 J_{10} + J_1^3 J_6 + J_1 J_{10}^3 &\equiv 0, \\ J_0^3 J_{10} + 6J_0 J_1 J_3 J_6 + 3J_0 J_1 J_{10}^2 + J_1 J_3^3 + J_3 J_6^3 + 7J_6^2 J_{10}^2 &\equiv 0. \end{aligned}$$

Now, modulo 11,

$$\sum_{n \geq 0} p(n)q^n = \frac{1}{E(q)} = \frac{(E(q)^3)^7}{E(q)^{22}} \equiv \frac{(E(q)^3)^7}{E(q^{11})^2} \equiv \frac{(J_0 + J_1 + J_3 + J_6 + J_{10})^7}{E(q^{11})^2}.$$

If we expand the right side and extract those terms in which the power of q is congruent to 6 modulo 11, we obtain

$$\sum_{n \geq 0} p(11n + 6)q^{11n+6} \equiv \frac{P}{E(q^{11})^2}$$

where

$$\begin{aligned} P &= 7J_0^6 J_6 + 10J_0^5 J_3^2 + J_0^4 J_1 J_6 J_{10} + 8J_0^3 J_1^3 J_3 + 2J_0^3 J_1 J_3^2 J_{10} + 8J_0^3 J_6^3 J_{10} \\ &+ 3J_0^2 J_1^2 J_3 J_6^2 + 3J_0^2 J_1^2 J_6 J_{10}^2 + 3J_0^2 J_3^2 J_6^2 J_{10} + 2J_0^2 J_3 J_6 J_{10}^3 + 10J_0^2 J_{10}^5 + 7J_0 J_1^6 \\ &+ J_0 J_1^4 J_3 J_{10} + 2J_0 J_1^2 J_3^2 J_6 + 3J_0 J_1^2 J_3^2 J_{10}^2 + J_0 J_1 J_3 J_6^4 + 2J_0 J_1 J_6^3 J_{10}^2 + J_0 J_3^4 J_6 J_{10} \\ &+ 8J_0 J_3^3 J_{10}^3 + 10J_1^5 J_6^2 + 2J_1^3 J_3 J_6^2 J_{10} + 8J_1^3 J_6 J_{10}^3 + 10J_1^2 J_3^5 + 8J_1 J_3^3 J_6^3 \\ &+ 3J_1 J_3^2 J_6^2 J_{10}^2 + J_1 J_3 J_6 J_{10}^4 + 7J_1 J_{10}^6 + 7J_3^6 J_{10} + 7J_3 J_6^6 + 10J_6^5 J_{10}^2 \\ &\equiv (7J_1 J_3 J_{10} + 5J_0^2 J_3 + 7J_1^3)(J_0^3 J_3 + J_0 J_1^3 + 6J_0 J_1 J_3 J_6 + 7J_1^2 J_6^2 + 3J_3 J_6^2 J_{10} + J_6 J_{10}^3) \\ &+ (7J_0^3 + 7J_0 J_1 J_{10} + 5J_6^2 J_{10})(J_0^3 J_6 + 7J_0^2 J_3^2 + 6J_0 J_1 J_6 J_{10} + J_1^3 J_3 + 3J_1 J_3^2 J_{10} + J_6^3 J_{10}) \\ &+ (7J_0 J_3 J_6 + 5J_0 J_{10}^2 + 7J_3^3)(3J_0 J_1^2 J_6 + 6J_0 J_3 J_6 J_{10} + J_0 J_{10}^3 + 7J_1^2 J_3^2 + J_1 J_6^3 + J_3^3 J_{10}) \\ &+ (5J_1^2 J_6 + 7J_3 J_6 J_{10} + 7J_{10}^3)(3J_0^2 J_3 J_6 + 7J_0^2 J_{10}^2 + J_0 J_3^3 + 6J_1 J_3 J_6 J_{10} + J_1^3 J_6 + J_1 J_{10}^3) \\ &+ (7J_0 J_1 J_6 + 5J_1 J_3^2 + 7J_6^3)(J_0^3 J_{10} + 6J_0 J_1 J_3 J_6 + 3J_0 J_1 J_{10}^2 + J_1 J_3^3 + J_3 J_6^3 + 7J_6^2 J_{10}^2) \\ &\equiv 0, \end{aligned}$$

and the proof is complete.

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